1. Problem Title & Source

Title: **Product of Digits for Product Pricing**

Source: **TCS NQT Previous Coding Questions**

2. Problem Statement (Short Summary)

A supermarket maintains a special pricing rule for its products.

Each product has a **number N printed on it**.

When the scanner reads the number:

- The **price of the item is calculated as the product of all digits of N**.

Your task is to **compute the product of the digits of the given number N**.

3. Examples

Example 1

Input

5244

Output

160

Explanation

Digits:

5 × 2 × 4 × 4

Result:

160

Example 2

Input

203

Output

0

Explanation

Digits:

2 × 0 × 3 = 0

4. Constraints

1 ≤ N ≤ 10^9

Observations

- Maximum digits ≈ **10**
- Straightforward digit extraction problem

5. Core Concept (Pattern)

Digit Manipulation

Important techniques:

modulo (%)

division (/)

6. Thought Process

Step 1 – Extract Digits

Use modulo operation:

digit = n % 10

Step 2 – Multiply Digits

Maintain:

product = 1

Multiply each digit.

Step 3 – Remove Last Digit

Use integer division:

n = n // 10

Repeat until number becomes **0**.

7. Visualization

Example

N = 5244

Steps

N	Digit	Product
5244	4	4
524	4	16
52	2	32
5	5	160

Final result

160

8. Pseudocode

read n

```
product = 1

while n > 0

   digit = n % 10
   product = product * digit
   n = n // 10

print product
```

9. Code Implementation

Python

```
n = int(input())

product = 1

while n > 0:

   digit = n % 10
   product *= digit
   n //= 10

print(product)
```

Java

```
import java.util.*;

class Main {

   public static void main(String[] args){

      Scanner sc = new Scanner(System.in);

      long n = sc.nextLong();

      long product = 1;
```

```
        while(n > 0){

            long digit = n % 10;

            product *= digit;

            n /= 10;
        }

        System.out.println(product);
    }
}
```

10. Time & Space Complexity

Time Complexity

O(log10 N)

Number of iterations equals **number of digits**.

Space Complexity

O(1)

Only a few variables are used.

11. Common Mistakes

1. Initializing product as **0**
2. Using floating division
3. Not updating the number after extracting digit
4. Handling negative numbers incorrectly

Correct initialization

product = 1

12. Detailed Dry Run

Input

203

Step	Number	Digit	Product
1	203	3	3

2	20	0	0
3	2	2	0

Final result

0

13. Real-Life Applications

Digit extraction appears in:

- Checksum calculations
- Barcode processing
- Digital signal processing
- Cryptography algorithms

14. Common Traps

1. Multiplying digits incorrectly
2. Missing last digit
3. Infinite loop if n not updated
4. Overflow for large numbers

15. Related Problems

Digit manipulation problems:

LC 258 Add Digits

LC 1281 Subtract Product and Sum of Digits

LC 66 Plus One

16. Alternate Approaches

Approach	Complexity	Notes
String conversion	O(d)	simple
Math digit extraction	O(d)	optimal

Where

d = number of digits

17. Why This Solution Works

The modulo operation extracts the **last digit** of the number.

Dividing by 10 removes the last digit.

Repeating this process ensures we process **every digit exactly once**.

18. Variations / Follow-Up Questions

Possible interview variations:

1. **Sum of digits**
2. **Reverse a number**
3. **Check Armstrong number**
4. **Count number of digits**